

Forms and CGI

Unit 3/5
BDBS205 (2025-2026)
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Forms

- Retrieving user inputs

```
<FORM METHOD = "POST" ACTION = "http://  
form.url.com/cgi-bin/cgiprogram">  
    ... Contents of the form ...  
</FORM>
```

- **Method** - GET/POST
- **Action** - URL Of the program to handle the http request (CGI or Java Servlet or JSP)

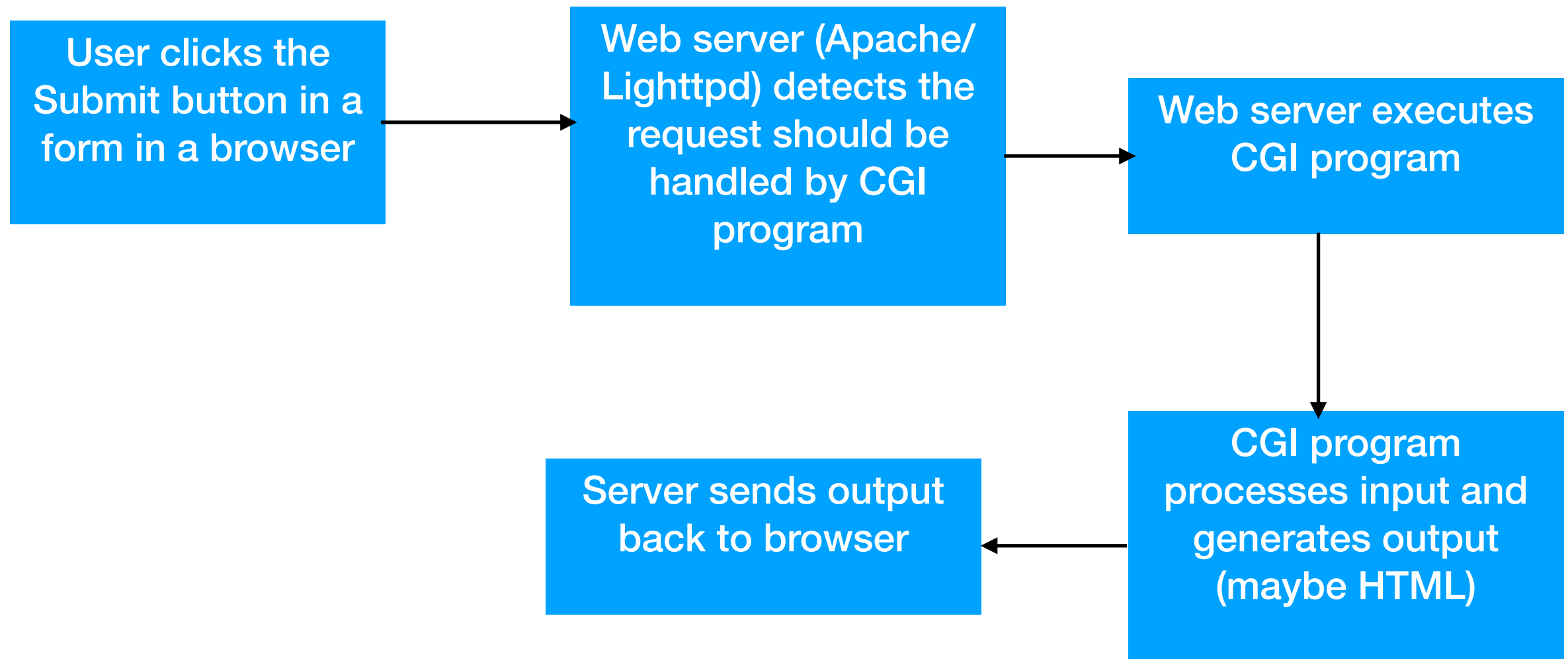
Form Elements

- `<INPUT TYPE="text" NAME="name_input" VALUE="default" MAXLENGTH=20 id="name1">`
- `<INPUT TYPE="checkbox" NAME="cbox" CHECKED id="chk1">`
- `<INPUT TYPE="radio" NAME="radio1" VALUE="1" id="rad">`
- `<INPUT TYPE="submit" VALUE="Done">`
- `<INPUT TYPE="reset" NAME="Reset">`
- `<INPUT TYPE="date" NAME="dob" id="dob">`
- `<INPUT TYPE="hidden" NAME="session_name" id="session_name" value="">`

CGI

- Common Gateway Interface
- Earliest Server-side programming
- Allows programs such as Perl, C, Python, Shell Script, Ruby to process form inputs and render dynamic HTML -
Language Independent
- Generates dynamic web content / responses to browser

Browser-CGI flow



Steps to run CGI script in web server

- Enable CGI in webserver :

```
sudo lighty-enable-mod cgi; systemctl restart lighttpd
```

- Ensure cgi.assign for python is present

```
vi /etc/lighttpd/conf-available/10-cgi.conf
```

```
cgi.assign = (  
  
    ".py" => "/usr/bin/python3"  
  
)
```

- Restart lighttpd
- Demo

CGI Environment Variables

```
•CONTENT_LENGTH:144

•CONTENT_TYPE:application/x-www-form-urlencoded

•DOCUMENT_ROOT:/var/www/html

•GATEWAY_INTERFACE:CGI/1.1

•HTTP_ACCEPT:text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

•HTTP_ACCEPT_ENCODING:gzip, deflate

•HTTP_ACCEPT_LANGUAGE:en-US,en;q=0.9

•HTTP_CACHE_CONTROL:max-age=0

•HTTP_CONNECTION:keep-alive

•HTTP_CONTENT_LENGTH:144

•HTTP_HOST:192.168.4.160:8080

•HTTP_ORIGIN:http://192.168.4.160:8080

•HTTP_REFERER:http://192.168.4.160:8080/form_example/form_example/form1.html

•HTTP_UPGRADE_INSECURE_REQUESTS:1

•HTTP_USER_AGENT:Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36

•HTTP_VIA:1.1 proxy.ibab.ac.in (squid/3.5.20)

•HTTP_X_FORWARDED_FOR:192.168.2.111

•LC_CTYPE:C.UTF-8

•QUERY_STRING:

•REDIRECT_STATUS:200

•REMOTE_ADDR:192.168.1.254

•REMOTE_PORT:50604

•REQUEST_METHOD:POST

•REQUEST_SCHEME:http

•REQUEST_URI:/cgi-bin/register.py

•SCRIPT_FILENAME:/var/www/html/cgi-bin/register.py

•SCRIPT_NAME:/cgi-bin/register.py

•SERVER_ADDR:192.168.4.160

SERVER_NAME:192.168.4.160

SERVER_PORT:8080

SERVER_PROTOCOL:HTTP/1.1
```


CGI Limitations

- Each Request - new Process (not thread)
- Scalability Issues (can handle only a few hundreds of requests)
- Replaced by PHP, Java Servlets, modern frameworks such as Flask, Node.js

PHP

- Developed as “Personal Home Page” by Rasmus Lerdorf
- Set of CGI scripts written in C to track site visits to home-page
- PHP1 (1994) — — — — — — — -> PHP8 (2020)
- Server side scripting, dynamic web-pages
- Replaced CGI
- Can also interact with back-end DB

Sample PHP:

```
<?php
```

```
    if ($_SERVER["REQUEST_METHOD"] == "POST") {  
        echo "<h2>Registration Successful</h2>";  
    }
```

```
?>
```

```
<html>
```

```
This is your usual html page
```

```
</html>
```